BT603 FINAL TERM 2020

Solved by ***ZAREEN FATIMA***

MCQS :

From Bull et al ,
kim et al ,
antifoams ,
jhonson euation
Fed batch
continous fermentation
Growth factor
buffers
Mcqs 40% were from midterms

QUESTIONS:

1. **What are anti foams 2marks**

Ans: Antifoams are surface active agents, reducing the surface tension in the foams and destabilizing protein films by (a) hydrophobic bridges between two surfaces, (b) displacement of the absorbed protein, and (c) rapid spreading on the surface of the film.

What is Replica plating ? 3 marks

Ans: Mutants can be detected by selecting or testing for an altered phenotype. Positive selection involves the selection of mutant cells and rejection of non-mutated cells. To identify mutants that have gained some physical trait, Replica plating is used for negative selection. Negative selection identifies mutants that have lost some physical trait - for example, auxotrophs that have nutritional requirements not possessed by the parent (nonmutated) cell.

2. **Important Antifoam names 3 marks**

Ans: The following compounds which meet most of ideal antifoam requirements (as discussed in previous module) have been found to be most suitable in different fermentation processes: 1. Alcohols; stearyl and octyl decanol. 2. Esters. 3. Fatty acids and derivatives, particularly glycerides, which include cottonseed oil, linseed oil, say-bean oil, olive oil, castor oil,

sunflower oil, rapeseed oil and cod liver oil. 4. Silicones. 5. Sulphonates. 6. Miscellaneous; Alkaterge C, oxazaline, poly-propylene glycol.

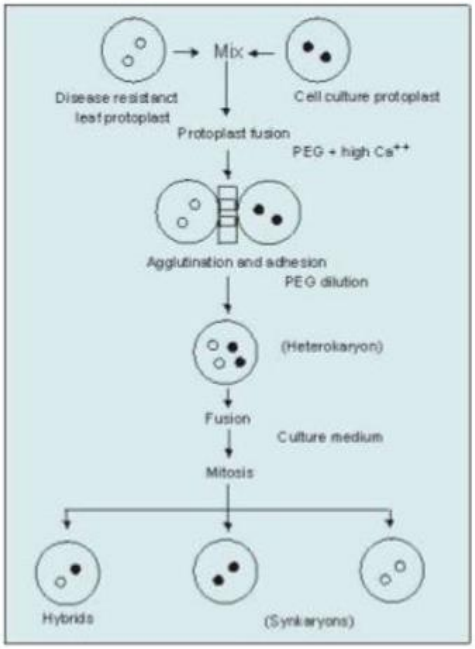
3. Note on AGAR SLANT

Ans: Storage on Agar Slants Cultures grown on agar slopes may be stored in a refrigerator (5°C) or a freezer (- 20°C) and sub-cultured at approximately 6-monthly intervals. The time of subculture may be extended to 0I year if the slopes are covered with sterile medicinal grade mineral oil.

4. Protoplast fusion method? 5 marks

Ans: Protoplasts are formed from bacteria, fungi, yeast and actinomycetes when dividing cells are caused to lose their cell walls. Fusion from mixed populations of protoplasts is greatly enhanced by the use of Polyethyleneglycol (PEG). The method has great industrial potential and has experimentally been used to achieve higher yields of antibiotics through fusion with protoplasts from different fungi. Protoplast fusion has been demonstrated as an efficient way to induce heterokaryon formation and recombination with high frequency.

- **Bacteria-grown plasmids are introduced into animal cells via fusion with bacterial protoplasts**
- **Bacterial cell walls are removed using lysozyme treatment**
- **Protoplasts fuse to the mammalian cells**
- **DNA purification not required**
- **may introduce bacterial components into recipient cells, cultures may be overgrown with bacteria**
- ***Bacteria are used to produce plasmids containing the human gene of interest. The cell walls are removed using lysozyme, resulting in the formation of protoplasts. The protoplasts are brought in contact with the mammalian cells, leading to protoplast fusion to mammalian cells, using Polyethylene glycol (PEG).***



5. stiometric equation for oxygen 5 marks

Ans: A number of workers have considered the overall stoichiometry of the conversion of oxygen, a source of carbon and a source of nitrogen into biomass and have used such relationships to predict the oxygen demand of a fermentation.

$$6.67CH_2O + 2.1O_2 = C_{3.92}H_{6.5}O_1 + 2.75CO_2 + 3.42H_2O$$

$$7.14CH_2 + 6.135O_2 = C_{3.92}H_{6.5}O_1 + 3.22CO_2 + 3.89H_2O$$

Terms used

$C_{3.92}H_{6.5}O_{1.94}$ is 100 g (dry weight) of yeast cells; CH_2O is carbohydrate

CH_2 is hydrocarbon

$$C = (A/Y) - B$$

A = Amount of oxygen for combustion of 1 g of substrate to CO_2, H_2O and NH_3, if nitrogen is present in the substrate

B = Amount of oxygen required for the combustion of 1 g cells to CO_2, H_2O and NH_3

Y = Cell yield (g cells g^{-1} substrate)

C = g oxygen consumed for the production of 1 g of cells

$$(dx/dt)/Y + mx + p = (dCO_2/dt) = -(dO_2/dt)RQ$$

x = biomass concentration

t = time

Y = g biomass g^{-1} carbon substrate

m = maintenance

p = allowance for antibiotic production

$$Y_{O/P} = (0.53/Y_{P/G}) - (0.6X/P) - 0.43$$

$Y_{O/P}$ = g oxygen consumed g^{-1} glucose
$Y_{P/G}$ = g sodium penicillin G produced g^{-1} glucose
X = g cells (dry weight) produced
P = g sodium penicillin G produced

$$Y_O = \{(32C + 8H - 16O)/YM\} - 1.58$$

Y_O = g oxygen consumed g^{-1} cells produced
Y = Cell yield (g cells g^{-1} substrate)
M = Molecular weight of the carbon source
C, H and O = Number of atoms of carbon, hydrogen and oxygen per molecule of carbon source

From Darlington's and Johnson's equations (in above mentioned Table) it may be seen that the production of 100 grams of biomass from hydrocarbon requires approximately three times the amount of oxygen to produce the same amount of biomass from carbohydrate.

6. Technique of microbial culture storage/preservation explain it? 10

Ans: The isolation of a suitable organism for a commercial process may be a long and very expensive procedure and it is therefore essential that it retains the desirable characteristics that led to its selection.

Also, the culture used to initiate an industrial fermentation must be viable and free from contamination. Thus, industrial cultures must be stored in such way as to eliminate genetic change, protect against contamination and retain viability.

 An organism may be kept viable by repeated subculture into fresh medium, but, at each cell division, there is a small probability of mutations occurring and because repeated sub-culture involves very many such divisions, there is a high probability that strain degeneration would occur.

 Also repeated sub-culture carries with it the risk of contamination. Thus, preservation techniques have been developed to maintain cultures in a state of 'suspended animation' by storing either at reduced temperature or in a dehydrated form.

 Storage at Reduced Temperature Storage on Agar Slants Cultures grown on agar slopes may be stored in a refrigerator (5°C) or a freezer (- 20°C) and sub-cultured at approximately 6-monthly intervals. The time of subculture may be extended to 0I year if the slopes are covered with sterile medicinal grade mineral oil.

Storage under Liquid Nitrogen The metabolic activities of micro-organisms may be reduced considerably by storage at the very low temperatures (-150° to -196°) which may be achieved using a liquid nitrogen refrigerator.

Fungi, bacteriophage, viruses, algae, yeasts, animal and plant cells and tissue cultures can all be successfully preserved by using this method. The technique involves growing a culture to the maximum stationary phase, resuspending the cells in a cryoprotective agent (such as 10% glycerol) and freezing the suspension in sealed ampoules before storage under liquid nitrogen.

 Storage in a Dehydrated Form Dried Cultures Dried soil cultures have been used widely for culture preservation, particularly for sporulating mycelial organisms. Moist, sterile soil may be inoculated with a culture and incubated for several days for some growth to occur and then allowed to dry

at room temperature for approximately 2 weeks. The dry soil may be stored in a dry atmosphere or, preferably, in a refrigerator. The technique has been used extensively for the storage of fungi and actinomycetes and Pridham et al. (1973) observed that of 1800 actinomycetes dried on soil about 50% were viable after 20-years storage.

 Storage in a Dehydrated Form Lyophilization During lyophilization (freeze-drying), a suspension of microbes is quickly frozen at temperatures ranging from -54°C to -72°C, and the water is removed by a high vacuum (sublimation). While under vacuum, the container is sealed by melting the glass with a hightemperature torch. The remaining powder like residue that contains the surviving microbes can be stored for years. The organisms can be revived at any time by hydration with a suitable liquid nutrient medium.

Lyophilization is very convenient for service culture collections because, once dried, the cultures need no further attention and the storage equipment (a refrigerator) is cheap and reliable.

Bt603 20feb 8am
Mcqs
_Mostly %age and formulas
_Lypholyzation temperature.
_Which type of fermentation need no oxygen

What is the difference between positive selection or negative selection mutant?

Ans: Mutants can be detected by selecting or testing for an altered phenotype. Positive selection involves the selection of mutant cells and rejection of nonmutated cells. To identify mutants that have gained some physical trait, Replica plating is used for negative selection. Negative selection identifies mutants that have lost some physical trait - for example, auxotrophs that have nutritional requirements not possessed by the parent (nonmutated) cell.

What is random mutant genesis?

Ans: Random Mutagenesis Early approaches to mutagenesis rely on methods which are entirely random in the mutations produced. Cells or organisms may be exposed to mutagens such as UV radiation or mutagenic chemicals, and mutants with desired characteristics are then selected. For Escherichia coli, mutants may be selected first by exposure to UV radiation, then plated onto agar medium. The colonies formed are then replica-plated, one in rich

medium, another in minimal medium, and mutants that have specific nutritional requirements can then be identified by their inability to grow in minimal medium. Similar procedures may be repeated with other types of cells and with different media for selection.

What is the role of inhibitor in fermentation medium?

Ans: In most cases the inhibitor is effective in increasing the yield of the desired product and reducing the yield of undesirable related products.

A number of studies have been made with potential chlorination inhibitors, e.g. bromide, to minimize chlortetracycline production during a tetracycline fermentation.

Inhibitors have also been used to affect cell-wall structure and increase the permeability for release of metabolites. The best example is the use of penicllin and surfactants in glutamic acid production.

What is protoplast?

Ans: Protoplasts are formed from bacteria, fungi, yeast and actinomycetes when dividing cells are caused to lose their cell walls.

What is organism accordingly bull man?

Ans: Bull et al. (1979) cited a criteria as being important in the choice of process organism:

1. The nutritional characteristics of the organism. It is frequently required that a process be carried out using a very cheap medium or a pre-determined one, e.g. the use of methanol as an energy source. These requirements may be met by the suitable design of the isolation medium.

2. The optimum temperature of the organism. The use of an organism having an optimum temperature above 40° considerably reduces the cooling costs of a large-scale fermentation and, therefore, the use of such a temperature in the isolation procedure may be beneficial.

3. The reaction of the organism with the equipment to be employed and the suitability of the organism to the type of process to be used.

4. The stability of the organism and its amenability to genetic manipulation.

5. The productivity of the organism, measured in its ability to convert substrate into product and to give a high yield of product per unit time.

6. The ease of product recovery from the culture.

Henry law?

Ans: Henry's Law describes the solubility of O2 in nutrient solution in relation to the O2 partial pressure in the gas phase

C*= P o/H

C* is the oxygen saturation concentration of the nutrient solution, P o is the partial pressure of the gas in the gas phase H is Henry's constant, which is specific for the gas and the liquid phase. Aeration with air 9 mg O2/L dissolves in water, with pure oxygen 43 mg O2/L.

Phynotypic characteristics of microorganisms?

Ans: • Microscopic morphology • Metabolic differences (Biochemical Testing) • Serology • Fatty acid analysis

Trace elements in fermentation media ?

Ans:The role of trace elements in medium formulation can be significant. Cultured cells normally require Fe, Zn, Cu, Se, Mn, Mo and V. These are often present as impurities in other media components. Cleveland et al. (1983) found that if the number of trace elements were increased, insulin, transferrin, albumin and liposomes were not needed in a serum-free hybridoma medium. They included Al, Ag, Ba, Br, Cd, Co, Cr, F, Ge, J, Rb, Zr, Si, Ni and Sn as well as those previously mentioned.

Equation of oxygen transfer from air bubble to liquid?

The rate of oxygen transfer from air bubble to the liquid phase may be described by the equation:

$$dC_L/dt = K_L a(C^* - C_L)$$

where C_L is the concentration of dissolved oxygen in the fermentation broth (mmoles dm^{-3}),

t is time (hours),

dC_L/dt is the change in oxygen concentration over a time period, i.e. the oxygen-transfer rate (mmoles O_2 dm^{-3} h^{-1}),

K_L is the mass transfer coefficient (cm h^{-1}),

a is the gas/liquid interface area per liquid volume (cm^2 cm^{-3}),

C^* is the saturated dissolved oxygen concentration (mmoles dm^{-3}).

define aeration 2 mark

Ans: Aeration refers to the process of introducing air to increase oxygen concentration in liquids. Aeration may be performed by bubbling air through the liquid, spraying the liquid into the air or agitation of the liquid to increase surface absorption.

Supplying oxygen to aerobic cells is a significant challenge. The problem is that oxygen is poorly soluble in water .

five enzyme name with inducer 3 marks

Ans:

Enzyme	Inducer	Micro-organism
α-Amylase	Starch Maltose	*Aspergillus spp.* *Bacillus subtilis*
Pullulanase	Maltose	*Aerobacter aerogenes*
α-Mannosidase	Yeast mannans	*Streptomyces griseus*
Penicillin acylase	Phenylacetic acid	*Escherichia coli*
Proteases	Various proteins	*Bacillus spp.* *Streptococcus spp.* *Streptomyces spp.* *Aspergillus spp.*
Cellulase	Cellulose	*Trichoderma viridae*
Pectinases	Pectin (beet pulp, apple pomace, citrus peel)	*Aspergillus spp.*
Nitralase	Isovaleronitrile	*Rhodococcus rhodochrous*

10 marks ka lyphozation process?

Ans: Storage in a Dehydrated Form Lyophilization During lyophilization (freeze-drying), a suspension of microbes is quickly frozen at temperatures ranging from -54°C to -72°C, and the water is removed by a high vacuum (sublimation). While under vacuum, the container is sealed by melting the glass with a hightemperature torch. The remaining powder like residue that contains the surviving microbes can be stored for years. The organisms can be revived at any time by hydration with a suitable liquid nutrient medium.

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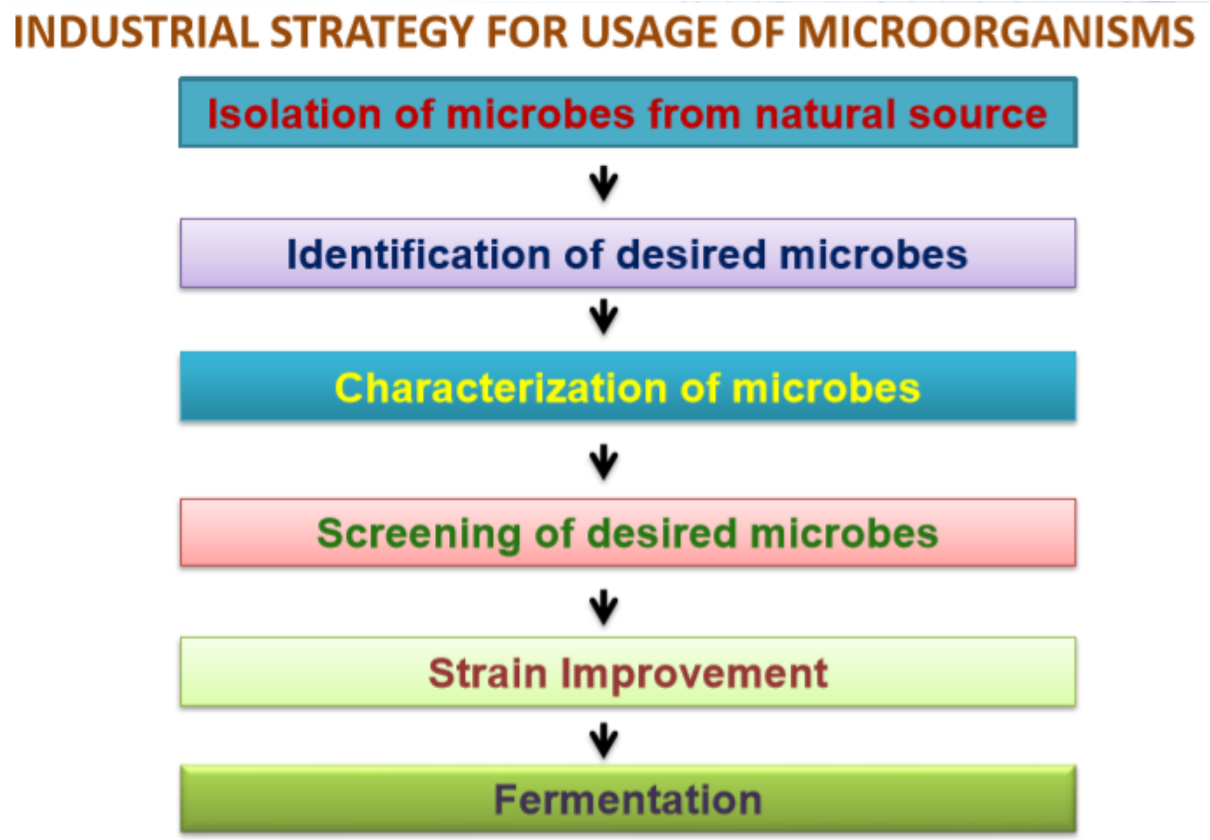
1: Two limitations of dynamic use of gassing out 2

Ans: A major limitation in the operation of the technique is the range over which the increase in dissolved oxygen concentration may be measured. It may be difficult to apply the technique during a fermentation which has an oxygen demand close to the supply capacity of the fermenter. Both the dynamic and static methods are also unsuitable for measuring KLa values in viscous systems.

2: What is pleotropic mutation 2

Ans: The selection of mutants is greatly facilitated by relying on the morphology of the mutants or on some selectivity in the medium. When morphological mutants are selected, it is in the hope that the desired mutation is pleotropic (i.e., a mutation in which change in one property is linked with a mutation in another character). The classic example of a pleotropic mutation is to be seen in the development of penicillin-yielding strains of Penicillium chrysogenum.

5: Industrial strategy of microorganisms show with a flow chart 5



Various method for identificaion of microbs?

Ans: Various techniques are employed to characterize and identify microorganisms:

– Phenotypic characteristics • Microscopic morphology • Metabolic differences (Biochemical Testing) • Serology • Fatty acid analysis

– Genotypic characteristics • Nucleic acid probes • DNA amplification • Ribotyping

Genotypic Characterization (Nucleic acid probes)

•Used to locate unique sequences Single-stranded DNA (or RNA)
•Generally labeled (radioactive or fluorescent) Complementary to the sequence of interest Observe and identify intact microorganisms
•Fluorescence in situ hybridization (FISH) Observe and identify samples
•DNA amplification by PCR Specific primers are used

•Sequencing rRNA genes (16S rRNA for bacteria) Evolutionarily highly conserved genes Variable regions are used to identify an organism. Particularly useful in identifying microbes that are difficult to culture.

3:Hall et al. (1973) have recognized five patterns of foaming in fermentations?5 marks

Ans: Hall et al. (1973) have recognized five patterns of foaming in fermentations: 1. Foaming remains at a constant level throughout the fermentation. Initially it is due to the medium and later due to microbial activity. 2. A steady fall in foaming during the early part of the fermentation, after which it remains constant. Initially it is due to the medium but there are no later effects caused by the micro-organism. 3. The foaming falls slightly in the early stages of the fermentation then rises. There are very slight effects caused by the medium but the major effects are due to microbial activity. 4. The fermentation has a low initial foaming capacity which rises. These effects are due solely to microbial activity. Media for Industrial Fermentations

5.A more complex foaming pattern during the fermentation which may be a combination of two or more of the previously described patterns.

4:If excessive foaming is encountered there are three ways

of approaching the problem?3 marks

Ans: If excessive foaming is encountered there are three ways of approaching the problem:

1. To try and avoid foam formation by using a defined medium and a modification of some of the physical parameters (pH, temperature, aeration and agitation). This assumes that the foam is due to a component in the medium and not a metabolite.

2. The foam is unavoidable and antifoam should be used. This is the more standard approach.

3. To use a mechanical foam breaker.

5:Bartholomew et al. (1950) represented the transfer of

oxygen from air to the cell, during a fermentation, as

occurring in a number of steps?3 marks

Ans: Bartholomew et al. (1950) represented the transfer of oxygen from air to the cell, during a fermentation, as occurring in a number of steps:

i. The transfer of oxygen from an air bubble into solution.
ii. The transfer of the dissolved oxygen through the fermentation medium to the microbial cell.

iii. The uptake of the dissolved oxygen by the cell.

These findings have been shown to be correct for nonviscous fermentations but it has been demonstrated that transfer may be limited by some other stages in certain highly viscous fermentations.

4 sources of protein?

Ans: Other proteinaceous nitrogen compounds serving as sources of amino acids include corn steep liquor, soya meal, peanut meal, cotton-seed meal.